2D Ideal MHD Solver on the Orszag-Tang Vortex

ME-7751 Final Project

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Outline

1. Introduction and Problem Motivation
2. Mathematical Model
3. Numerical Model
4. Validation and Results
5. Discussion of Results
6. Conclusions

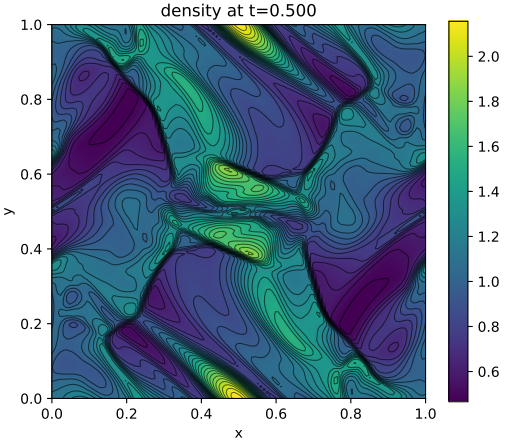
Why MHD

- Magnetohydrodynamics couples fluid flow and magnetic field through the Lorentz force.
- Valid when plasma is collisional and magnetic Reynolds number is large: solar wind, tokamaks, astrophysical jets, magnetospheres.
- New physics vs Euler: Alfven and fast/slow magnetosonic waves, the Maxwell stress, and the constraint $\nabla \cdot B = 0$.

Project goal. Build a 2D ideal MHD finite-volume solver in Python and validate it on the Orszag-Tang vortex.

Why Orszag-Tang

- Canonical 2D MHD benchmark.
- All-sinusoidal ICs on $[0, 1]^2$, periodic on every side.
- No geometry, no density jump, no outflow BCs.
- If the result is wrong, it is the solver, not the setup.
- Reference: Londrillo & Del Zanna 2000, Athena test suite.



Ideal MHD in conservative form

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0, \quad U = (\rho,\ \rho\mathbf{v},\ \mathbf{B},\ E)^T$$

with total energy $E = p/(\gamma - 1) + \frac{1}{2}\rho|\mathbf{v}|^2 + \frac{1}{2}|\mathbf{B}|^2$ and total pressure $p_T = p + \frac{1}{2}|\mathbf{B}|^2$.

- Eight conserved variables; all three velocity and field components are carried in 2D.
- Seven-wave fan: two fast magnetosonic, two Alfven, two slow magnetosonic, one entropy.
- Ideal-gas closure, $\gamma = 5/3$.

The divergence constraint

- Exact ideal MHD preserves $\nabla \cdot B = 0$ analytically.
- A piecewise-constant Godunov update does not: centered finite-differences produce spurious divergence at cell centers.
- Consequences: parallel-to-B Lorentz force, loss of momentum conservation, eventual blow-up.
- Remedies: constrained transport, Powell source terms, or hyperbolic GLM cleaning.
- **This project uses GLM**: simpler to drop into an unstaggered FVM code, only adds one scalar ψ.

Scheme at a glance

- 2D structured Cartesian grid, cell-centered state, two ghost layers.
- MUSCL linear reconstruction on primitives with minmod limiter (MC, van Leer optional).
- HLLD Riemann solver (Miyoshi & Kusano 2005) at every interface; degenerates to HLLC for $B = 0$.
- GLM psi/Bx coupling handled inside the Riemann solver; parabolic damping applied per RK stage.
- SSP-RK2 time integration, CFL $= 0.3$.

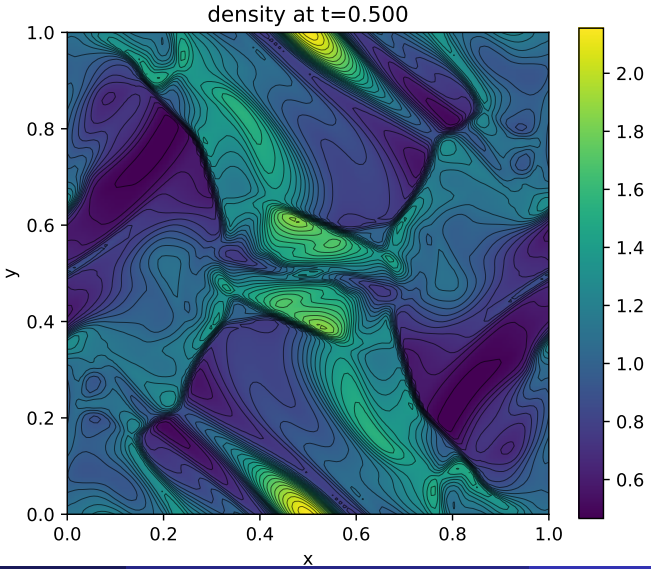
HLLD wave pattern

- Five-wave approximation to the full seven-wave MHD fan.
- Bounding speeds S_L, S_R: Davis-Einfeldt.
- Entropy speed S_M from jump conditions at the contact wave.
- Two starred states and two double-starred states (across the rotational Alfven discontinuities).
- Positivity-preserving for physical inputs.

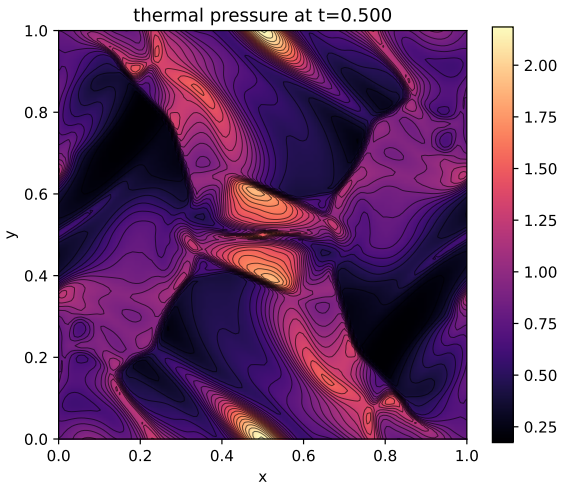
L	L^*	L^{**}	R^{**}	R^*	R
S_L		S_L^*		S_M	S_R

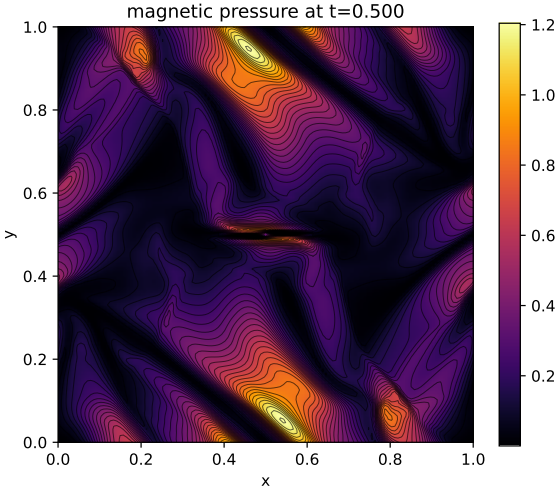
Wave regions across the x-face.

Density at t = 0.5, 128^2



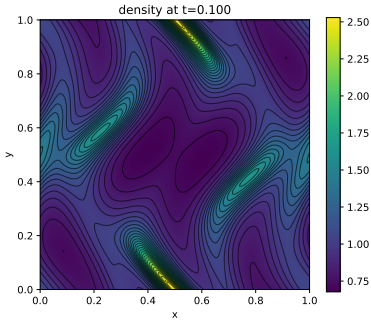
Thermal and magnetic pressure, t = 0.5



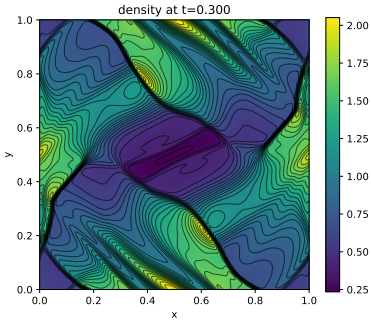


- Thermal pressure tracks density with extra shock heating near the current sheet.
- Magnetic pressure peaks at flux pile-up in the four current-sheet arms.

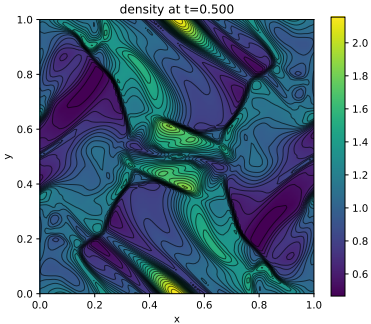
Formation of the current sheet



t = 0.1



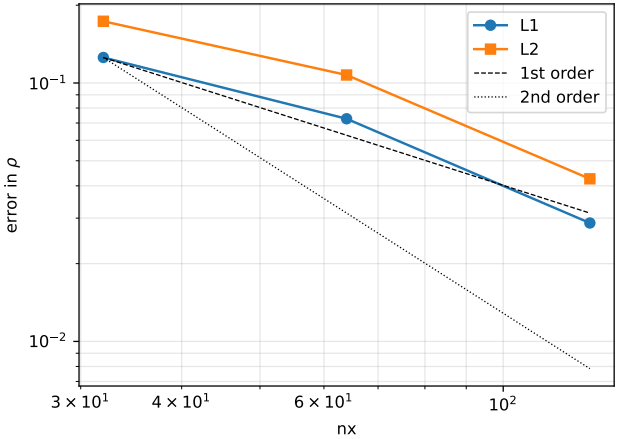
t = 0.3



t = 0.5

- t = 0.1: still smooth, the vortices are just forming.
- t = 0.3: diagonal shocks have formed; bow-tie compression starts.
- t = 0.5: fully-developed current sheet and bounding shocks.

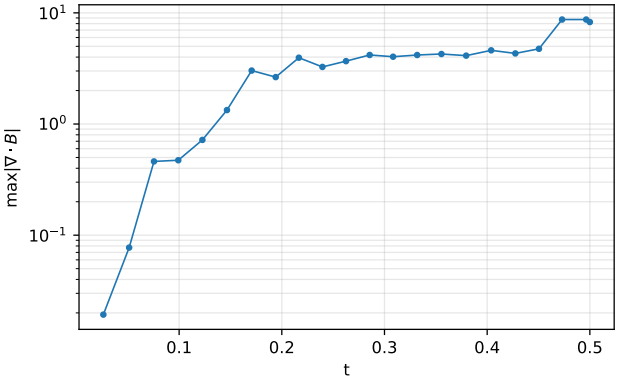
Convergence in density



nx	L^1	L^2
32	1.26e-1	1.73e-1
64	7.27e-2	1.07e-1
128	2.87e-2	4.26e-2

- $32 \to 64$: rate ≈ 0.79.
- $64 \to 128$: rate ≈ 1.33.
- Between 1st and 2nd order: expected for minmod on a shock-dominated flow.

Divergence control and conservation



- GLM bounds $\max|\nabla \cdot B|$ once shocks form.
- Cell-centered centered differences are a pessimistic diagnostic; the GLM-natural measure is smaller.
- Mass, momentum, energy conserved to machine precision ($\sim 10^{-16}$ drift per step) – see `run.log`.

Accuracy vs limiter choice

- Observed density extrema at 128^2: $\rho_{\min} \approx 0.47$, $\rho_{\max} \approx 2.16$.
- Athena++ with MC limiter at the same resolution: tighter extrema ($\rho_{\min} \sim 0.1$, $\rho_{\max} \sim 3.5$).
- Gap is almost entirely the limiter: minmod is the most diffusive TVD limiter and smears current sheets fastest.
- MC or van Leer would close the gap but at a cost in robustness for low-beta cells.

Why between first and second order

- MUSCL is second-order in smooth regions.
- Any TVD limiter drops to first order at local extrema and at shocks.
- Orszag-Tang at $t = 0.5$ is shock-dominated, so the global rate sits between 1 and 2.
- Observed: ≈ 0.8 at coarse resolutions (shock-dominated), ≈ 1.3 at fine (smooth regions start to dominate).

Limitations

- Vectorized NumPy: 128^2 runs in $\sim 7\,\mathrm{s}$; 256^2 is the practical ceiling on a laptop.
- No constrained-transport option – GLM is simpler but less strict about $\nabla \cdot B$.
- 2D Cartesian only; no adaptive refinement.
- Reconstruction is linear (MUSCL) – no WENO, no higher-order time integration.

Conclusions

- Built a 2D ideal MHD solver from scratch: Godunov FVM, MUSCL-minmod, HLLD, GLM, SSP-RK2.
- Orszag-Tang at 128^2 reproduces the canonical density/pressure pattern at $t = 0.5$.
- Mass, momentum, energy conserved to machine precision; $\max |\nabla \cdot B|$ bounded by GLM.
- Convergence rate between 1 and 2, consistent with MUSCL-minmod on a shock-dominated flow.

Natural extensions. Constrained transport; resistive MHD; 3D; AMR; Numba/JAX compilation.

References

- Miyoshi & Kusano, *J. Comput. Phys.* 208, 315 (2005) – HLLD.
- Dedner et al., *J. Comput. Phys.* 175, 645 (2002) – GLM cleaning.
- Londrillo & Del Zanna, *ApJ* 530, 508 (2000) – Orszag-Tang reference.
- Powell et al., *J. Comput. Phys.* 154, 284 (1999) – 8-wave formulation.
- Brio & Wu, *J. Comput. Phys.* 75, 400 (1988) – 1D shock-tube reference.
- Toro, *Riemann Solvers and Numerical Methods for Fluid Dynamics* (2009) – MUSCL, HLL family.
- Versteeg & Malalasekera, *An Intro. to CFD: The Finite Volume Method* (2007) – FVM fundamentals.
- Chen, *Intro. to Plasma Physics* (2016); Freidberg, *Ideal MHD* (2014) – plasma/MHD physics.
- Athena test suite: `https://www.astro.princeton.edu/~jstone/Athena/tests/orszag-tang/`.